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SOFT SWITCHING IN SWITCHED INERTANCE HYDRAULIC CIRCUITS

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ABSTRACT

Switched Inertance Hydraulic Systems (SIHS) use inductive, capacitive, and switching elements to boost or buck a pressure from a source to a load in an ideally lossless manner. Real SIHS circuits suffer a variety of energy losses, with throttling of flow during transitions of the high-speed valve resulting in 44% of overall losses. These throttling energy losses can be mitigated by applying the analog of zero-voltage-switching, a soft switching strategy, adopted from power electronics. In the soft switching circuit, the flow that would otherwise be throttled across the transitioning valve is stored in a capacitive element and bypassed through check valves in parallel with the switching valves. To evaluate the effectiveness of soft switching in a boost converter SIHS, a lumped parameter model was constructed. The model demonstrates that soft switching can improve the efficiency of the circuit up to 42% and extend the power delivery capabilities of the circuit by 76%.

INTRODUCTION

Switched inertance hydraulic systems (SIHS) are switch mode fluid power circuits analogous to buck and boost voltage converters in the field of power electronics. Buck and boost converters provide an ideally lossless method for converting an input DC voltage to an output DC voltage. Converter circuits produce a DC output voltage may be lower than the input voltage in the case of a buck converter, or greater than to input voltage in the case of a boost converter [1]. Hydraulic buck and boost converters behave as DC-DC pressure transformers and can be applied in place of throttling valve control in circuits where multiple actuators are powered from a constant pressure rail.

Fluid power SIHS circuits were proposed by Brown in 1984 as an efficient alternative to lossy throttling valve control [2]. SIHS circuits utilize fluid mass in a long conduit as an inertive device that has the same function as the inductor in the power electronic circuit.

In application, switch mode fluid power circuits experience pipe flow resistance losses and orifice throttling losses during valve transitions. Rannow and Li identified that valve transitions accounted for 60% of the overall losses in the virtually variable displacement pump configuration that they simulated [3]. They discussed mitigating these losses by placing a small spring piston accumulator in parallel with the tank side switching valve. This piston accumulator was a passive device through the first half of the piston travel. The second half of the piston travel was restricted by an active locking mechanism which released the piston at the onset of the tank valve opening transition. This locking mechanism was critical in the function of the proposed soft switching method. A pressure based locking piston accumulator for this purpose was designed by Van de Ven that eliminated the need for external control over the locking mechanism [4]. This design was experimentally validated by Beckstrand and Van de Ven [5]. Beckstrand found that the devices performed as expected under a narrow range of experimental conditions, but the implementation of complicated geometry of the locking device posed a challenge to the device over wider operating ranges.

Power electronic converters also experience switching and conduction losses [6]. To mitigate switching losses, 'zero voltage switching' techniques are applied. This approach seeks to eliminate the voltage drop, and hence losses, across a transitioning switch. Henze proposed adding capacitors and

diodes in parallel with the transistors in a boost converter to achieve zero voltage switching [6]. In his circuit, a switch overlap is implemented, during which time the circuit enters a resonant phase due to the inductance in the circuit and the added capacitors. The resonance of the circuit charges or discharges the capacitors to a voltage that is equal to either the load or low-supply voltage. Diodes in parallel with the capacitors prevent voltage overshoot during this resonant phase.

The hydraulic boost converter:

In this work, the hydraulic boost converter, depicted in Figure 1, is used as a basis for efficiency improvement. This circuit is referred to as the ‘baseline’ circuit in contrast to the soft switched circuit presented later.

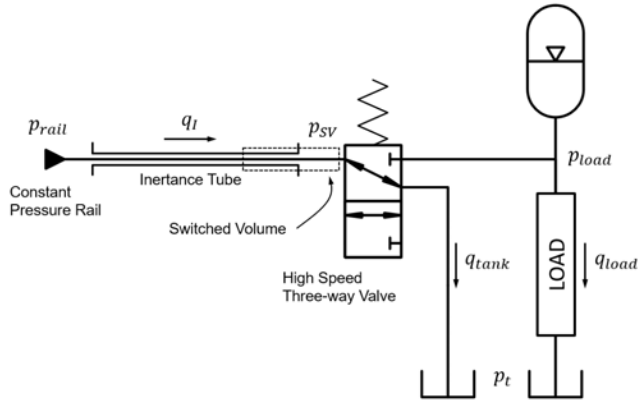


Figure 1: Baseline boost converter hydraulic circuit.

The duty cycle of the boost converter is the ratio of time that flow is directed to tank over the switching period. An ideal boost converter behaves according to the following relationship:

$$p_{load} = \frac{p_{rail}}{1 - D} \quad (1)$$

where p_{load} is the load pressure, p_{rail} is the supply pressure and D is the duty cycle. Similarly, and by conservation of power, a lossless boost converter would have the following relationship between mean rail flow rate and load flow rate:

$$q_{rail} = \frac{q_{load}}{1 - D} \quad (2)$$

where q_{load} is the mean load flow rate and q_{rail} is the mean rail flow rate.

Work conducted by Johnston et al. explains baseline boost converter operation and presents simulation results for a boost converter operating over a range of load flow conditions [7]. De Negri et al. describes load pressure behavior given a linear inertance tube resistance model with analytical equations [8]. These models do not capture valve losses, but agreed well with experimental results when an oversized and relatively fast transitioning valve was used. Pan et al. developed a frequency domain model of the entire boost converter system that captured

inertance tube resistance, but assumed linear valve orifice resistance and instantaneous transitions [9]. These models were modified in a following paper, in which the inertance tube was modeled in the frequency domain and the rest of the system was modeled in the time domain [10]. This mixed time-frequency approach, which was proposed by Manhartsguber [11], allows simulation of wave delay in the inertance tube as well as simulation of non-linear behavior such as orifice resistance and variable orifice area in the rest of the system.

In this work, the inertance tube is modeled utilizing a lumped parameter time domain model, which is valid for conduit lengths less than 4% of the switching wavelength [12,13]. These models offer a simplified, but valid look at system performance in the absence of the complex transients introduced by wave delay effects for boost converters with short inertance tubes. A high speed valve model, based on a crank-slider actuated valve capable of 100Hz switching and 0.8ms transition, is included to represent real valve losses [14].

In the next section, a soft switching hydraulic boost converter is proposed to mitigate valve transition losses. Description of the circuit modeling procedure is next, followed by description of the simulation procedure. In the next section, the simulation results are discussed. The final section makes concluding remarks.

SOFT SWITCHING

Soft switching in the context of this work is a switching methodology rather than a particular device. The concepts herein are adapted from the zero-voltage switching methodology applied to electronic boost converters by Henze [6]. The goal of soft switching is to reduce switching valve transition losses by equalizing the pressure across the valve during periods of transition.

Theory of Operation:

An idealized switch mode circuit with instantaneous valve transition and an inviscid working fluid is lossless. However, a physical system experiences finite valve transition time, which results in forcing flow through a partially opened orifice. These throttling events result in a large pressure drop, according to the orifice equation. Power losses due to throttling flow across an orifice are expressed in Eq. (3):

$$P_{loss} = \Delta p Q = \Delta p C_d A \sqrt{\frac{2\Delta p}{\rho}} \quad (3)$$

where A is the instantaneous valve area. From Eq. (3), if the pressure differential across the orifice can be reduced to zero, throttling losses can be eliminated.

A soft switched boost converter has an additional spring loaded accumulator and check valves applied in parallel with the switching valves, as shown in Figure 2. These passive elements modulate the pressure in the “switched volume” at the outlet of

the inertance tube, adjacent to the switching valves. The pressure differential, and hence losses across a transitioning valve, is minimized by charging this volume to the load pressure or discharging this volume to the tank pressure prior to those respective valve opening events. Similarly, the capacitive element buffers the pressure in the switched volume against sudden change during valve closing events. The circuit achieves soft switching utilizing the described passive devices and a valve overlap, during which time both load and tank valve are closed. This period allows the capacitive elements to charge and discharge prior to valve opening transitions. The 3-way valve in Figure 1 is separated into two 2-way valves in Figure 2 for clarity of discussion.

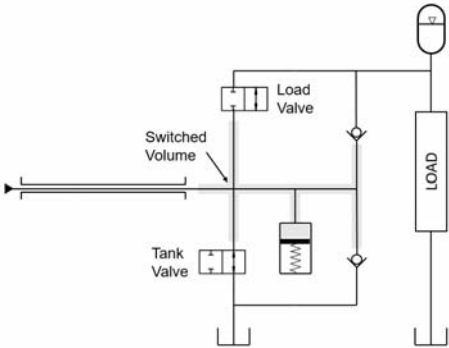


Figure 2: Soft switching boost converter utilizing spring loaded capacitive elements.

Sequence of Operation:

Figure 3a	The cycle is initialized by flowing to tank, building velocity and thus kinetic energy and momentum in the fluid in the inertance tube. The load accumulator is large enough that it can supply flow to the load for the duration of time that the pressure supply is shunted to tank.
Figure 3b	At time DT , the tank valve is closed. The momentum of the flow in the inertance tube charges the switched volume accumulator (SVA). Once the pressure in the switched volume reaches the load pressure, flow is allowed through the load check valve, equalizing the pressure across the load side valve.
Figure 3c	The load side valve is opened across a minimal pressure differential. The adverse pressure gradient causes the flow in the inertance tube to decelerate and eventually reverse.
Figure 3d	The flow direction in the inertance tube is now towards the pressure supply. The load side valve closes and the momentum of the flow decompresses the switched volume. Once the switched volume reaches tank pressure, the tank side check valve opens, equalizing pressure across the tank side valve. The tank side valve now opens across a minimal pressure differential, initiating the next switching cycle.

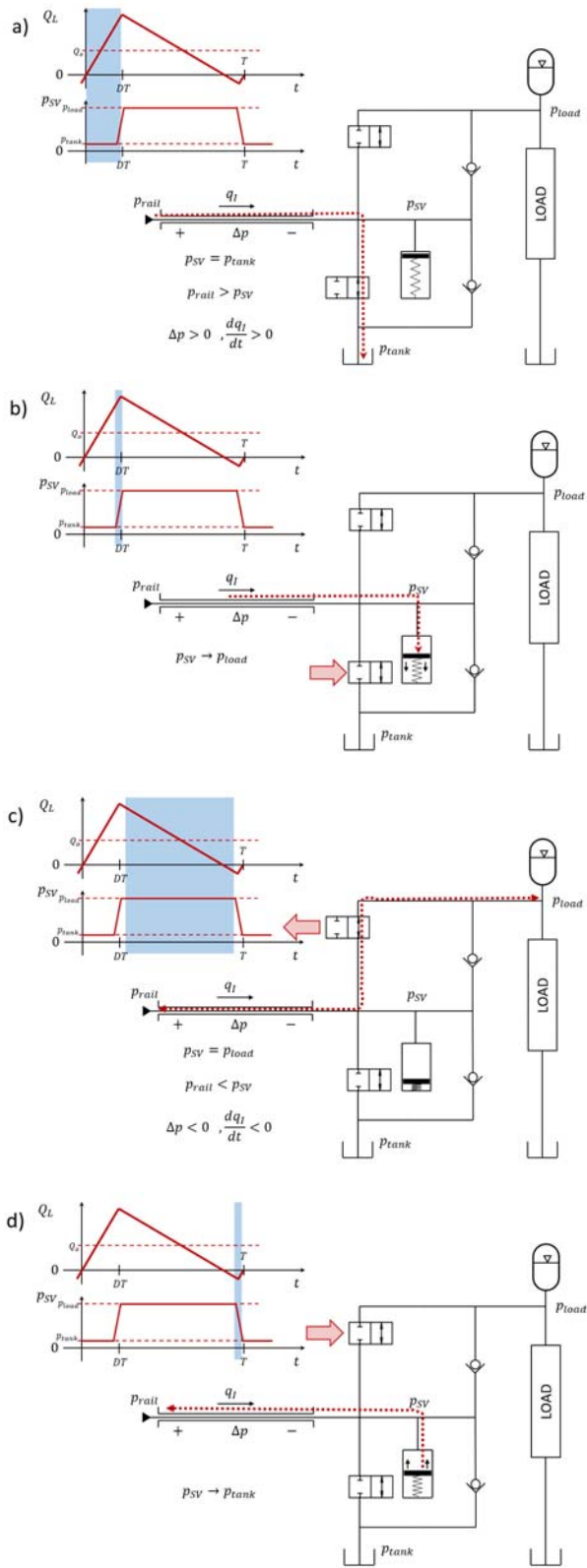


Figure 3: Soft switched boost converter sequence of operation

Calculating capacitance for soft switching

The soft switching boost converter is a direct hydraulic analogy of the zero voltage resonant transition circuit topology proposed by Henze and shown in Figure 4 [6]. Henze presents several analytical relationships that are used to size the capacitive elements with respect to electrical current flow and switch overlap time.

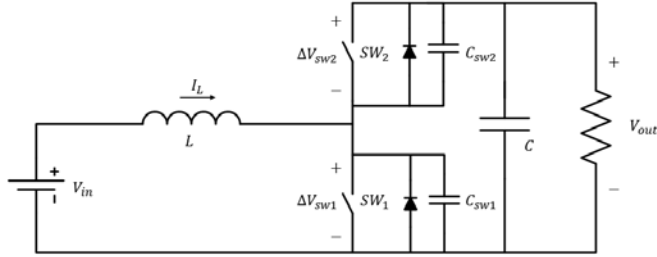


Figure 4: Zero voltage switching boost converter. Image from[6].

When both switches are open, the soft switched boost converter in Figure 4 reduces to a resonant inductive-capacitive (L-C) circuit. Henze proposes the relationship in Eq. (4) to ensure that the capacitors are fully charged during this period:

$$T_{OL} > \frac{2C_{sw}\Delta V_{sw}}{|I_L|} \quad (4)$$

where T_{OL} is the length of time that both switches are open, C_{sw} is the capacitance of one of the capacitors in parallel with the switch, ΔV_{sw} is the voltage differential across the switch and I_L is the current in the inductor at the time of the switching event. It is assumed that T_{OL} is small relative to the switching period, T and that the switch transition time is small relative to both T_{OL} and T . An inequality appears in Eq. (4) due to the parallel diode allowing current to bypass the capacitor once fully charged or discharged. The limiting condition for determining the minimum switching overlap time is the electric current level at the SW_1 switch opening transition. At this time, the current is at a low magnitude negative value, a similar state as the hydraulic boost converter in Figure 3d.

Power electronics have the benefit of short transition times, often several orders of magnitude shorter than the switching period. In order to minimize the switching period in hydraulic switch mode circuits, the transition time of the valve is on the order of 5% of the switching period. Due to the relatively long transition period of the hydraulic valve, a substantial capacitance is required to buffer the switched volume pressure during the transition. Eq. (4) assumes negligible discharge of the capacitance during the transition period, and does not establish a minimum capacitive value. Assuming a linear switching valve orifice area, it is reasonable to add 50% of the transition time to the overlap time in Eq. (4) to account for capacitor discharge during the relatively long switch transition time. With this

considered, a capacitance value that will buffer the switched volume during transition but also fully charge during the valve overlap can be calculated directly. Re-stating in hydraulic terms, Eq. (4) becomes:

$$T_{OL} + \frac{1}{2}T_{TR} = \frac{C_{SV}\Delta p}{q_I} \quad (5)$$

where T_{TR} is the valve transition time, C_{SV} is the hydraulic capacitance of the switched volume, Δp is the pressure differential between the supply and tank when calculating the tank side transition or supply and load when calculating the load side transition. q_I is the inertance tube flow rate at the time of transition.

Equations (6) and (7) show hydraulic capacitance values for a spring-piston accumulator and a compressible fluid volume, respectively.

$$C = \frac{A_p^2}{k_s} \quad (6)$$

$$C = \frac{V_c}{\beta} \quad (7)$$

where C is the hydraulic capacitance, A_p is the area of the piston, k_s is the spring rate, V_c is the volume of the compressible fluid and β is the bulk modulus of the fluid.

The tank closing transition event in Figure 3b takes place at peak flow, and the load closing transition event in Figure 3d at minimum inertance tube flow. At moderate duty cycles, the pressure differential Δp term can be considered approximately equal prior to load side or tank side transitions. Assuming equal transition time and overlap time, the ideal switched volume capacitance for each of these transition events is proportional to the flow rate magnitude at the time of the transition event according Eq. (5). This means that the capacitive value of the SVA will be a compromise between optimal switching to load or to tank, lying somewhere between the two values predicted by Eq. (5) when either peak or minimum flow is considered.

To address the asymmetry in capacitance required for effective switching at the high flow and low flow transition events, the piston travel can be limited such that the capacitance of the accumulator goes to zero above a fixed pressure. At switched volume pressures greater than this cutoff pressure, further piston travel is restricted and the capacitance of the switched volume is only due to the compliance of the fluid in the inertance tube and the SVA cylinder, per Eq. (7). Doing this allows a large capacitance at the low pressure/high flow conditions that correspond to the tank valve transition and smaller capacitance at the high pressure/low flow conditions that occur at load valve transition.

MODEL

Inertance Tube Model

Ideally, the inertance tube is purely an inertive element. However, viscous flow through a conduit results in friction losses and compliance of the fluid within the conduit results in capacitive effects. Both of these non-ideal terms are distributed spatially throughout a physical system. The physical system experiences wave delay in the transmission of pressure signals from one end of the conduit to the other, which travel at the sonic velocity. In this work, a lumped parameter approximation is applied to the inertance tube in order to better illustrate the soft switching concept by simplifying the complex system dynamics that occur in wave delay models. To maintain validity of the lumped parameter modeling approach, the inertance tube length is set to 4% of the switching period wavelength, calculated as:

$$\lambda = cT \quad (8)$$

where λ is the switching period wavelength, c is the sonic velocity in the fluid. Lumped parameter systems convey information from one computational node to the next instantaneously, but application of the inertance tube length, $l_{tube} = 0.04\lambda$ standard minimizes the effect of wave delay to the point that it can be accurately neglected.

The 'medium transmission line' modeling approximation is adapted from the field of power electronics to represent the inertive, resistive, and capacitive elements of the inertance tube [13]. The inertance tube in Figure 5 is modeled as series inertive and resistive elements, with parallel capacitive elements at either end. Each capacitive element represents the compliance of one half of the total fluid volume contained in the tube.

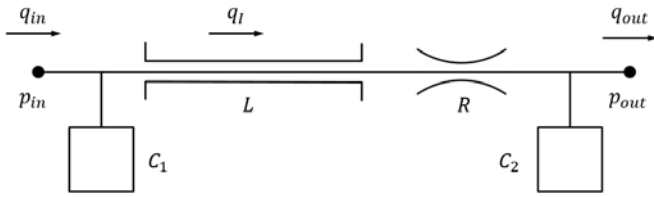


Figure 5: Lumped parameter inertance tube 'medium line' model

In Figure 5, L is the inertance of the fluid in the inertance tube, calculated as:

$$L = \frac{\rho l_{tube}}{A} \quad (9)$$

where ρ is the density of the hydraulic fluid, and A is the cross sectional area of the inertance tube. R is the resistance of the tube, where in the laminar case:

$$R = \frac{8\pi\mu l_{tube}}{A^2} \quad (10)$$

where μ is the kinematic viscosity of the hydraulic fluid. C_1 and C_2 are the capacitance values of the oil volumes at either end of the inertance tube model:

$$C_1 = C_2 = \frac{1}{2} \frac{V_{tube}}{\beta} \quad (11)$$

where V_{tube} is the volume of the inertance tube, and β is the bulk modulus of the hydraulic fluid.

The flow rate in the inertive element is a function of the inertance in the tube, and the pressure difference applied across the tube, less the pressure drop due to resistive losses:

$$p_{in} - p_{out} = L \frac{dq_l}{dt} + q_l R \quad (12)$$

where p_{in} is the inertance tube inlet pressure, p_{out} is the outlet pressure, and q_l is the flow rate through the inertive lumped element.

The pressures in the inlet and outlet volumes are given by the following equations:

$$\frac{dp_{in}}{dt} = \frac{1}{C_1} (q_{in} - q_l) \quad (13)$$

$$\frac{dp_{out}}{dt} = \frac{1}{C_2} (q_l - q_{out}) \quad (14)$$

In the boost converter simulation, the inertance tube inlet is connected to the rail pressure, which is modeled as a constant pressure boundary condition.

Switching Valve Model

The modeled boost converter is driven by a simulated crank-slider three-way valve switching at 100Hz [14]. This valve model is simplified, approximating the load and tank sides as separate valves with port orifices of $3 \times 10^{-5} \text{m}^2$. The modeled valve area profile is linear and transitions in 0.8ms.

The flow rates through the load and tank branches of the three-way valve are calculated by the orifice equation:

$$q_v = C_d A_v(t) \sqrt{\frac{2\Delta p_v}{\rho}} \quad (15)$$

where q_v is the flow rate through the valve, C_d is the discharge coefficient, $A_v(t)$ is the instantaneous valve open area, and Δp_v is the pressure differential across the valve.

Load Model

The load on the circuit is modeled as a constant volumetric flow rate sink.

Check Valve Model

The check valves are modeled as ideal elements that instantaneously opening once the cracking pressure has been exceeded. Once the check valve is open, it is modeled as an orifice:

$$q_c = \begin{cases} 0 & \text{if } p_1 - p_2 < p_{crack} \\ C_d A_c \sqrt{\frac{2(p_1 - p_2)}{\rho}} & \text{if } p_1 - p_2 > p_{crack} \end{cases} \quad (16)$$

where q_c is the cracking pressure of the check valve, A_c is the orifice area of the check valve, p_{crack} is the check valve cracking pressure, and p_1 and p_2 are the pressures at the inlet and outlet of the check valve, respectively. The check valve areas in the model are set to 75% of the switching valve orifice area, to ensure that the check valves close once the switching valve has opened. The cracking pressure of both check valves was set to 0.10MPa.

Gas Charged Load Accumulator Model

The load accumulator is charged with nitrogen and modeled as an ideal gas undergoing isentropic expansion and compression:

$$p_{acc} V_{gas}^\gamma = p_{charge} V_{acc}^\gamma \quad (17)$$

where p_{acc} is the instantaneous accumulator pressure, p_{charge} is the accumulator charge pressure, V_{acc} is the accumulator volume, γ is the specific heat ratio of the charge gas. V_{gas} is the instantaneous volume of the charge gas.

Spring-Piston Switched Volume Accumulator Model

The spring-piston accumulator applied at the switched volume is modeled as a purely capacitive device:

$$\frac{dp_{SV}}{dt} = \frac{1}{C_{SP}} q_{SP} \quad (18)$$

where C_{SP} is the capacitance of the SVA, per Eq. (6), and q_{SP} is the flow rate into the spring piston accumulator.

Circuit Efficiency

The efficiency of the boost converter is calculated by dividing the power delivered to the load over a cycle by the power supplied by the rail, averaged over a cycle:

$$\eta = \int_t^{t+T} \frac{q_{load}(\tau)(p_{load}(\tau) - p_{tank})}{q_I(\tau)p_{rail}} d\tau \quad (19)$$

SIMULATION RESULTS

The simulated boost converter circuits were developed around a 6MPa rail pressure and nominal load flow rate of 0.1L/s. In each simulation, the inertance tube is initialized at zero flow and at rail pressure. The load is initialized at ambient pressure and given flow rate. The simulation proceeds until the load pressure achieves steady state for 20 switching cycles, defined as less than 0.1% pressure variation between consecutive cycles. Simulated duty cycles are $0.1 \leq D \leq 0.9$. Duty cycles less than 0.1 and greater than 0.9 experience insufficient valve opening for proper operation due to the valve transition time. The hydraulic fluid properties used in the simulation are listed in Table 1. Note that a constant bulk modulus is assumed to simplify the model and maintain focus on illustrating the role of soft switching.

Table 1: Physical properties of the simulated hydraulic fluid.

Hydraulic Fluid Properties		
Density	874	kg/m^3
Bulk Modulus	1.40	GPa
Kinematic viscosity	0.0404	$Pa \cdot s$
Entrained Air Fraction	0	%

The baseline boost converter was simulated with an inertance tube length of 0.506m. The variation in inertance tube flow over a cycle is a function of the inertance value of the tube, with longer and thinner tubes having higher inertance and correspondingly lower variation in flow [8]. The inertance tube diameter for the baseline circuit was selected such that continuous positive flow rate through the inertance tube is achieved over the range of duty cycles at 0.1L/s load flow rate. Ensuring only positive flow rate through the inertance tube prevents the soft switching effect of discharging the switched volume prior to tank valve transition as occurs in Figure 3d. The De Negri et al. analytical model, which includes laminar pipe friction [8] predicts this value to be 3.3mm. The simulations presented in this work include switching valve resistance and finite transition time, which are not captured by the De Negri et al. model. The simulations demonstrate that an inertance tube diameter of 3.9mm results in continuous positive flow in the inertance tube throughout the boost converter operational range when considering valve losses. Figure 6 presents the simulated baseline boost converter inertance tube flow rates over a range of duty cycles for an inertance tube length $l_t = 0.506m$ and inertance tube diameter $d_t = 3.9mm$.

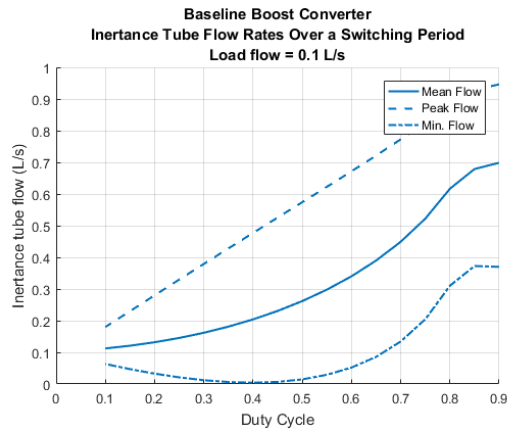


Figure 6: Simulation results of inertance tube mean, peak and minimum flow rates over a switching period vs. duty cycle for the baseline boost converter.

Next, a single switching cycle during cyclic steady state operation of the baseline boost converter is analyzed. The parameters of the simulated baseline and soft switched boost converters are outlined in Table 2 and the performance of the converters is outlined in Table 3. Full open valve losses are the power losses that the flow would experience when passing through a fully opened valve. Transition losses are the valve losses in excess of the full open valve losses. Friction losses are the laminar flow losses in the inertance tube.

Table 2: Physical parameters of the simulated boost converters

Boost Converter Simulation Parameters			
	Baseline	Soft Switched	
Switching Speed	100		Hz
Valve Overlap	0.3		ms
Duty	0.7		
Valve Orifice Area	3×10^{-5}		m^2
Valve Transition Time	0.8		ms
Inertance tube length	0.506		m
Inertance tube diameter	3.9		mm
Load Flow rate	0.1		L/s
Switched volume accumulator:			
Piston Diameter	N/A	3.6	mm
Piston Travel	N/A	3.0	mm
Spring Rate	N/A	10000	N/m

Table 3: Baseline boost converter performance metrics

Boost Converter Performance, $D = 0.7$			
	Baseline	Soft Switched	
Efficiency, η	29.71	60.51	%
Load Pressure	7.97	11.65	Mpa
Transition Losses	8.13	1.39	J/cycle
Full Open Valve Losses	1.85	0.70	J/cycle
Friction Losses	8.62	5.45	J/cycle

The baseline boost converter experiences pressure spikes at $t = 0.35$ s and $t = 0.65$ s in Figure 7. The pressure spikes occur when the flow in the inertance tube is forced through the transitioning valve orifices. These spikes result in strong adverse pressure gradients in the inertance tube, which decelerate the inertance tube flow at the same temporal locations, as shown in Figure 7.

The losses over a cycle are shown in Figure 8. The most significant loss event occurs at the load valve opening transition at 0.35 s $< t < 0.4$ s. The major pressure spike that occurs prior to this event results in a large pressure differential across the valve and hence major throttling losses.

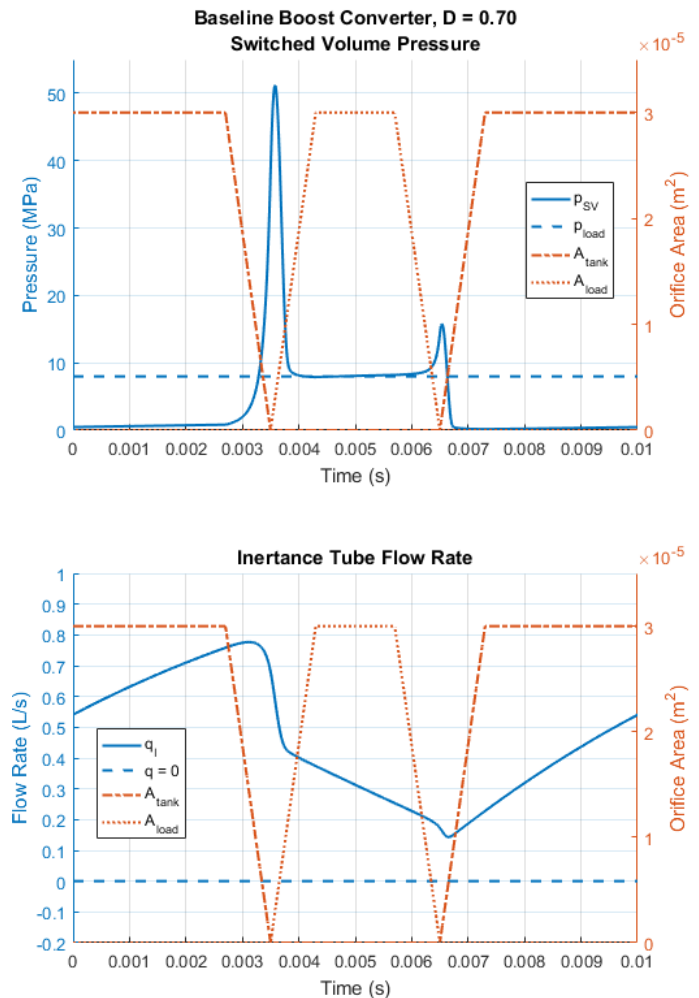


Figure 7: Switched volume pressure inertance tube flow in a baseline boost converter, $D = 0.7$.

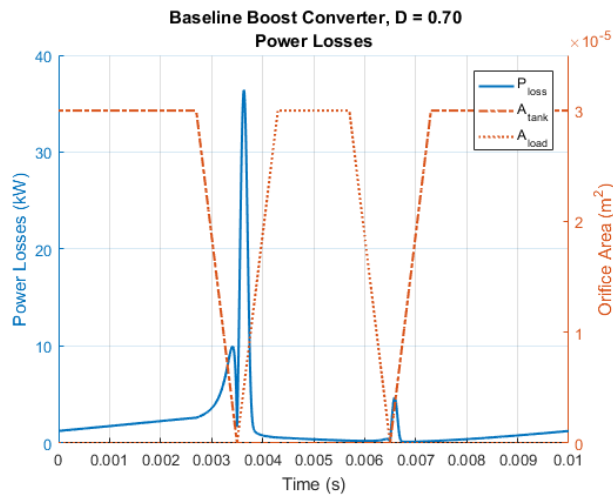


Figure 8: Power Losses over a switching cycle in a baseline boost converter, $D = 0.7$.

The parameters of the simulated soft switched system are identical to those of the baseline system with the exception of the added SVA and check valves. The SVA piston hits a stop once the internal pressure reaches 3MPa, effectively disabling the accumulator until the pressure goes below this level. The soft switched converter exhibits 82% lower transition losses relative to the baseline system, reducing per cycle transition losses from 8.13 J/cycle to 1.39J/cycle. Figure 9 shows the soft switched system dynamics, on the same scales as Figure 7.

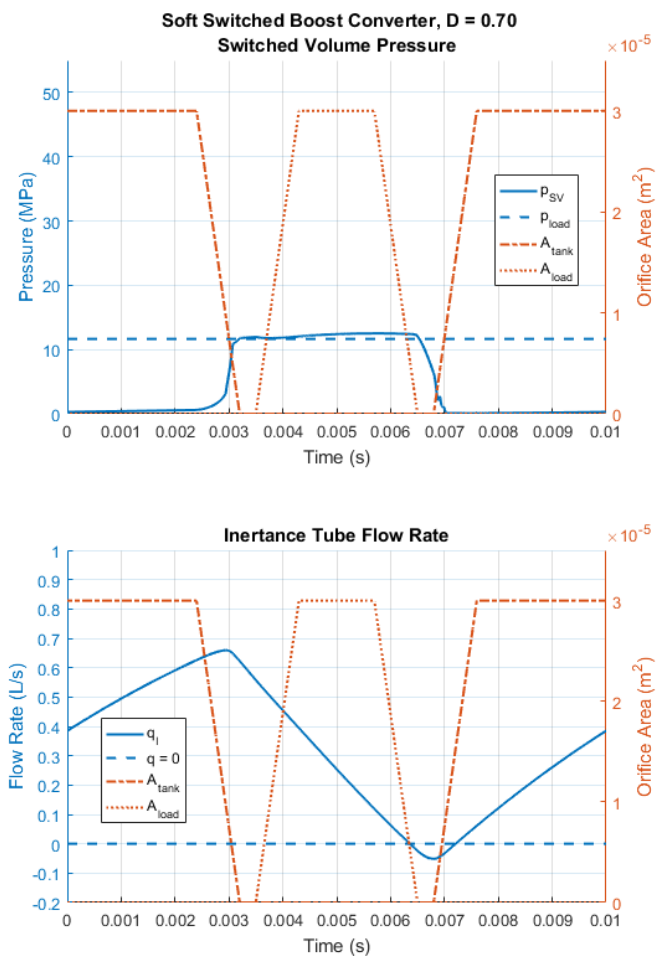


Figure 9: Switched volume pressure and inertance tube flow in a soft switched boost converter, $D = 0.7$.

In the soft switched boost converter, the SVA absorbs flow during the tank closing event, slowing the rate at which the pressure rises in the switched volume. At time $t = 0.0029$ s in Figure 9, the piston hits the bottom of its travel, negating its capacitive function. After this time, the pressure ramps up quickly as a result of the much lower capacitive value of the inertance tube fluid compliance. The pressure spike prior to the load valve opening event is mitigated by the check valve in parallel with the load valve, which provides an alternate flow path once the switched volume pressure is greater than the load pressure. The pressure in the switched volume falls during the overlap after the load valve closes. At time $t = 0.0068$ s, the accumulator piston begins traveling, discharging fully at time $t = 0.007$ s.

Comparing the flow rate plots in Figures 7 and 9, the mean flow in the soft switched boost converter inertance tube is lower than the mean flow rate in the baseline boost converter. This is due to the baseline circuit having greater flow through the inertance tube than that predicted by Eq. (2). This excess flow, termed ‘flow loss,’ is related to the efficiency of the circuit. Less

efficient circuits experience greater flow loss at the same duty cycle and load flow rate [9]. The increased mean flowrate through the inertance tube of the baseline boost converter results in greater friction and fully open valve losses relative to the soft switched boost converter.

The power loss of the soft switched boost converter over a cycle is compared to that of the baseline converter in Figure 10. In this figure it is clear that the major transition loss events are eliminated or reduced in the soft switching circuit.

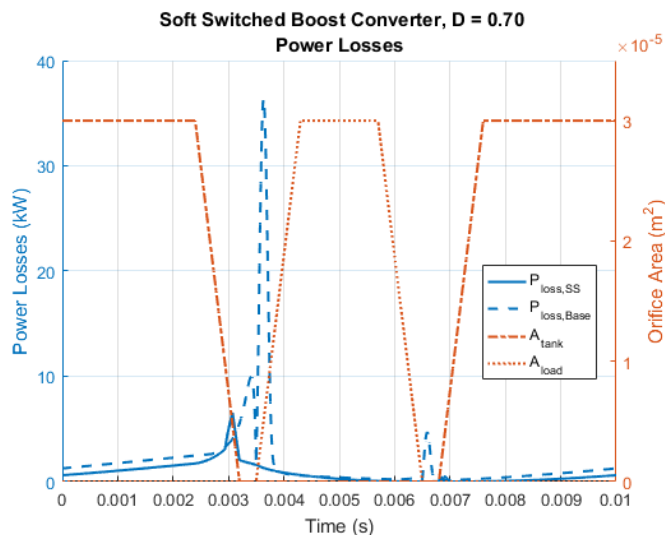


Figure 10: Power losses over a switching cycle in a soft switched boost converter compared to power losses in a baseline boost converter, $D = 0.7$.

Figures 7-10 focus on the cyclic steady state circuit behavior at a duty cycle of 0.7. Figure 11 compares the efficiency of the two systems over a range of duty cycles.

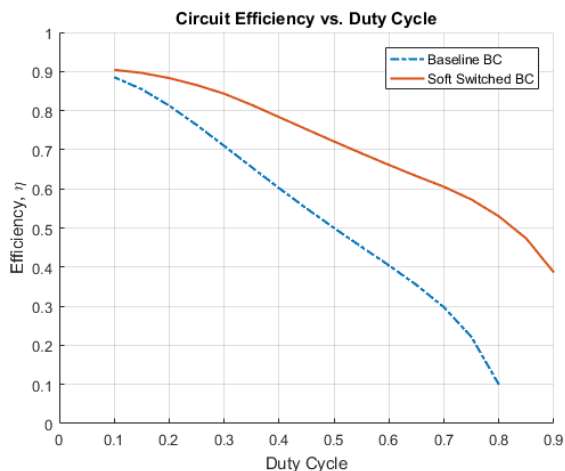


Figure 11: Efficiency vs. duty cycle for a baseline and soft switched circuit. Load flow rate 0.1L/s.

The efficiency results for the baseline boost converter are similar to the results predicted and experimentally verified by

De Negri et al [8]. The soft switched boost converter offers efficiency improvements over the range of operation relative to the baseline circuit which result in greater load pressure for a given duty than the baseline circuit, as shown in Figure 12.

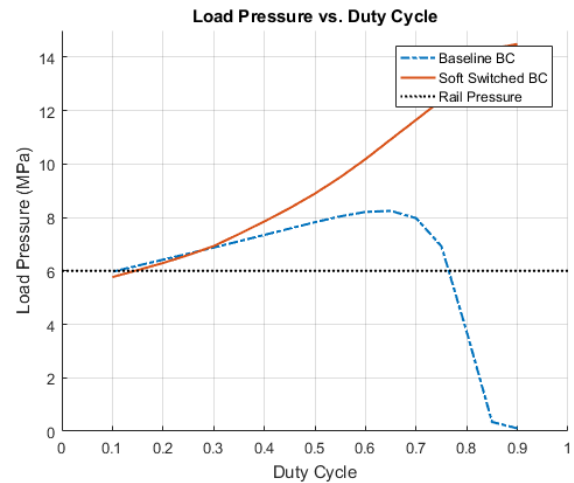


Figure 12: Load pressure vs. duty cycle for a baseline and soft switched circuit. Load flow rate 0.1L/s

The increased load pressure that the soft switched circuit is able to deliver results in greater power delivery across the range of duty cycles as shown in Figure 13. The soft switched boost converter is able to deliver nearly double the maximum power of the baseline boost converter. At duty cycles $D > 0.7$, the load pressure delivered by the baseline circuit drops due to circuit losses. This drop in load pressure delivery is seen where the baseline boost converter load power bends back toward lower power values and lower efficiency.



Figure 13: Efficiency vs. load power delivery for a baseline and soft switched circuit. Load flow rate 0.1L/s

DISCUSSION

The SVA was designed for a fixed load flow rate of 0.1L/s, with the boost converter operating at a duty cycle $D = 0.7$. This SVA, combined with the check valves resulted in performance improvements across the range of operation. The system would likely benefit from an optimization process to select the best SVA capacitance value and cutoff pressure considering the range of duty cycles and various load flow rates.

The SVA presented in the work is small and has a high spring rate. Such a spring-piston accumulator would be difficult to produce. Calculating the required capacitance value for the SVA via Eq. (5) results in values of $7.0 \times 10^{-14} \text{ m}^3/\text{Pa}$ at the high flow, tank closing transition and $1.2 \times 10^{-14} \text{ m}^3/\text{Pa}$ at the low flow, load closing transition.

Studies of distributed parameter switch mode converter circuits have found that the length of the inertance tube, switching frequency and duty cycle should have the following relationships for maximum efficiency [7,9]:

$$l_{tube} = \frac{TDc}{2} \quad 0 < D < 0.5 \quad (20)$$

$$l_{tube} = \frac{T(1-D)c}{2} \quad 0.5 \leq D < 1$$

These formulas result in an optimal inertance tube length of 25% of the maximum switching speed wavelength, where the switching wavelength is calculated by Eq. (8). 100Hz switching and sonic velocity of 1265m/s result in an inertance tube length of 3.16m. This inertance tube length is outside of the validity of lumped parameter modeling, but the outlet capacitive value of such a tube can be approximated by Eq. (11), given a tube diameter.

An inertance tube of 3.16m in length and having diameter of 7.5mm has an outlet capacitive value of $4.9 \times 10^{-14} \text{ m}^3/\text{Pa}$. This indicates that the compliance of the fluid in the inertance tube alone is enough to achieve soft switching, eliminating the need for the SVA when optimal length inertance tubes are applied. The diameter of the inertance tube affects the inertive, capacitive and resistive values of the conduit, all of which have a bearing on circuit performance. For this reason, the best inertance tube diameter would be discovered through optimization.

The check valves that allow flow to bypass the switching valves prior to valve opening events must operate quickly relative to the switching period. Ideally these valves operate instantaneously, but a physical system will rely on valves that open in a fraction of a millisecond. Research into small high speed plate type hydraulic check valves has demonstrated them to be effective in a 300Hz pumping application with potential to operate at 400-500 Hz [15]. Reed style check valves with low moving mass have demonstrated high speed potential in

automotive applications and are currently being studied for hydraulic applications [16].

This work focuses on soft switching in a boost converter. Henze presents a similar soft switching buck converter that could readily be adapted to a hydraulic circuit using the approach presented in this paper. The soft switching buck converter also relies on passive elements without the need for external, mechanical control.

CONCLUSION

A soft switching boost converter circuit was presented. This circuit aims to reduce the losses due to flow throttling through a transitioning switching valve. The circuit adds check valves in parallel with the switching valves and a spring piston accumulator at the outlet of the inertance tube. All elements added to the circuit are passive, and do not require external control for operation. The non-soft switched and soft switched boost converter were simulated using a lumped parameter approach. The lumped parameter models remain valid as the length of the inertance tube was restricted to 4% of the switching wavelength, minimizing wave delay effects.

The results of the soft switched boost converter were compared to a boost converter of equal inertance tube dimensions lacking soft switching strategies. The soft switching boost converter provided up to 42% gains in efficiency and 77% improvement in load pressure delivery for a fixed load flowrate of 0.1L/s. When the length of the inertance tube is increased close to an optimal length, the inherent capacitance of the fluid in the inertance tube is sufficient to achieve soft switching, eliminating the need for an accumulator in the switched volume.

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